CL 420: Biochemical Engineering

Course Project

Final Report and Presentation Instructions

This document sets out the requirements for Stage 2 of the course project, which carries 70% of the project grade. Teams should continue with the product and host organism registered in Stage 1. Stage 2 consists of a written report and a team presentation followed by Q&A. **The report and presentation should reflect the understanding of the concepts covered in the course.**

1. Report Format

- **Length:** Maximum 12 pages including the title page
- **Cover page:** project title, names and roll numbers of all team members
- **Font:** Times New Roman, 12 pt.
- **Spacing:** 1.5
- **Calculations:** show the key working in the main text (equation, values used, result with units).
- **Data sources:** cite a source for every kinetic, yield and process parameter.
- **References:** Follow any standard format consistently.

2. Report Content

The report should address the following sections:

2.1 Product and Process Definition

- Product description, its application, target market and scale.
- Choice of host organism or cell line, with justification.
- Product characteristics, and critical quality attributes (CQAs).

2.2 Growth and Product formation kinetics

- Nature of product formation: growth associated, non growth associated or mixed, with justification.
- Key metabolic pathways; substrates, nutrients and by-products.
- Mode of operation (batch, fed-batch or continuous) and the reason for the choice.

2.3 Bioreactor Design

- Bioreactor type and scale, and its suitability for the organism (oxygen demand, shear sensitivity, broth viscosity).
- Reactor sizing, flow rates, batch time and other key design parameters.
- Operating conditions: temperature, pH, dissolved oxygen, agitation and aeration, with a basis for each value.
- Inoculum development, media composition, feeding strategy and process duration.
- Monitoring and control (PAT): parameters measured online and offline, and those used for control.

2.4 Engineering Calculations

- Material balance over the bioreactor
- Growth and product formation kinetics
- Estimation of Reactor volume, flow rates, concentrations, batch time and the predicted titer, rate and yield.
- Yield coefficients $Y_{x/s}$ and $Y_{p/s}$.
- Oxygen transfer: oxygen uptake rate (OUR), required k_La, and the agitation and aeration needed to achieve it.

- A summary table of key parameters and results for sections 2.3 and 2.4 is recommended at the end.

2.5 Downstream Processing and Rough Economics (in brief).

- Sequence of major recovery and purification steps, preferably as a block flow diagram.
- Main cost drivers: media and feed, reactor time, oxygen and aeration, purification.
- Give rough estimates per batch or per kg of product.

3. Final Presentation

Presentation slides should contain highlights of report as well as reflect the concepts covered in the course.

Each team will present using 7 slides: 1 introduction slide followed by 6 slides for the actual content.

Each student will present 2 slides for a total of 2 minutes to present their slides.

The presentation will be followed by a Q&A session.

Time slots for presentation will be announced shortly.

Presentation guidelines

Do's	Don'ts
• Ensure your slides are clearly visible from the back of the room. • Keep slides concise, with one main idea per slide. • Arrive on time (arrive 5 min before your slot; you are welcome to attend other slots as well). • Ask questions to other teams. Insightful questions will be awarded bonus points.	• Do not exceed the time limit of 2 minutes per student • Do not exceed 7 slides • Do not arrive late. • Do not disturb others. • Do not read directly from the slides or from your mobile phone or from notes.

4. Evaluation Criteria

- Clear and concise report and presentation covering the points in section 2.
- Satisfactory answers in the Q&A session.
- Similarity score and AI content in the report.

5. Submission

- Submit the report as a single PDF on Moodle. Submit the presentation as a separate PDF. Announcements will be made soon on how to submit. Late submissions will not be entertained.